



The iconoclastic Paracelsus, an alchemist and chemist, was one of the greatest scientists of the 16th century, and advocated the use of minerals in healing, but only in tightly controlled dosages.

experience and hard work.” And again, “What a doctor needs is not eloquence or knowledge of language and of books, but profound knowledge of nature and her works.” He also paid great attention to the exact dosage, saying that “it depends only on the dose whether a poison is a poison or not.”

As a result, Paracelsus was an influential force in the future development of chemistry, conventional medicine, herbal medicine, and homeopathy. Though he is known as the “father of chemistry,” he also explored alchemy, which concerned itself with the transmutation of base materials to gold, and the search for immortal life. Paracelsus also revived interest in the Doctrine of Signatures—the ancient theory that held that a plant’s appearance indicated the ailments it would treat—and affirmed the value of locally grown medicinal herbs over expensive imported specimens.

Culpeper & Printed Herbals

Paracelsus’ advocacy of local herbs was later fiercely espoused by Nicholas Culpeper (1616–1654). The frontispiece to his *The English Physitian* contains the memorable words: “Containing a Compleat Method of Physick, whereby a Man

may preserve his Body in Health, or Cure himself, being Sick, for three pence Charge, with such things only as grow in England, they being most fit for English Bodies.”

Wounded during the English Civil War fighting for the Commonwealth, Culpeper championed the needs of the ordinary people who could afford neither the services of a doctor nor the expensive imported herbs and formulations that doctors generally prescribed. Drawing to some degree on Dioscorides, Arabian physicians, and Paracelsus, Culpeper developed a medical system that blended astrology and sound personal experience of the therapeutic uses of local plants. His herbal became an instant “bestseller” and appeared in many subsequent editions. The first herbal published in North America, in 1700, was an edition of his herbal.

While the popularity of *The English Physitian* was notable, other herbals also found a place in households. The development of the printing press in the 15th century brought herbal medicine into homes on a wide scale. Texts such as Dioscorides’ *De Materia Medica* were printed for the first time, and throughout Europe herbals were published and ran through many editions.

Deadly Cures 1700–1900

By the end of the 16th century, Paracelsus had become the figurehead of the new chemical medicine. However, where he had insisted upon caution in the use of metallic poisons—mercury, antimony, and arsenic—the new medical thinkers were not so inhibited. Larger and larger doses of the purgative known as calomel (mercurous chloride, Hg_2Cl_2) were given to those suffering from syphilis and many other diseases. The treatment was very often worse than the illness, with some patients dying and many more suffering from the long-term consequences of mercury poisoning.

Hippocrates’ saying “Desperate cases need the most desperate remedies” was taken very literally, as is evident in the incredible excess of purging and bleeding that developed over the next three centuries in Europe and North America. These practices reached a peak in the “heroic” medicine of the early 19th century. Its leading proponent, Dr. Benjamin Rush (1745–1813), maintained that only bloodletting and calomel were required in medical practice. His position was obviously extreme, but it is clear that in this new climate herbal medicines were becoming increasingly irrelevant.

The New Rationalism

Along with the new emphasis on chemical cures, modern medicine came to look askance at the notion of the “vital force.” Up until the end of the 16th century, nearly all medical traditions had been based on the concept of working with nature, with the body’s healing capacities, which could be supported and strengthened with appropriate medicinal herbs. In traditional Chinese medicine, *qi* is the primal energy that maintains life and health. In Ayurveda, it is *prana*, and